

FH 1

Vincent Wang

January 2026

1 Definitions

Let X be a finite set and V be the vector space with basis $\{e_x\}_{x \in X}$. Let $G : X$ by $ge_x = e_{gx}$ and extending linearly to V . Identify e_x with $\delta_x \in \text{Hom}(X, \mathbb{C})$. Show that the definitions agree.

First,

$$\begin{aligned} g\left(\sum a_x e_x + \sum b_x e_x\right) &= g\left(\sum (a_x + b_x) e_x\right) = \sum (a_x + b_x) e_{gx} \\ &= \sum a_x e_{gx} + \sum b_x e_{gx} = g\left(\sum a_x e_x\right) + g\left(\sum b_x e_x\right), \end{aligned}$$

$$g\left(\lambda \sum a_x e_x\right) = g\left(\sum \lambda a_x e_x\right) = \sum \lambda a_x e_{gx} = \lambda \sum a_x e_{gx} = \lambda g\left(\sum a_x e_x\right),$$

so g is indeed linear.

Second, $1e_x = e_{1x} = e_x$, so 1 acts as the identity of V .

Third,

$$g(h(e_x)) = g(e_{hx}) = e_{g(hx)} = e_{(gh)x} = (gh)(e_x),$$

so extending to V , we have that G indeed acts on V .

Suppose G acts on V . Define $\varphi(g)(v) = gv$. I claim that $\varphi : G \rightarrow L(V)$ is a homomorphism. Note that $g^{-1}(gv) = v$ for all $v \in V$, so $g : V \rightarrow V$ is invertible. g is assumed linear, so $g \in L(V)$. Further,

$$\varphi(gh)(v) = (gh)(v) = g(h(v)) = \varphi(g)(h(v)) = \varphi(g)(\varphi(h)(v)),$$

so $\varphi(gh) = \varphi(g)\varphi(h)$.

Suppose $G : V$. Define the left module action of $G\mathbb{C}$ on V by $g \cdot v = gv$.

Indeed,

$$\begin{aligned}
1 \cdot v &= 1v = v \\
g \cdot (v + w) &= g(v + w) = gv + gw = g \cdot v + g \cdot w \\
(g + h) \cdot v &= (g + h)(v) = gv + hv = g \cdot v + h \cdot v
\end{aligned}$$

We check associativity:

$$\begin{aligned}
\left(\sum_x a_x g_x \sum_y b_y g_y \right) \cdot v &= \left(\sum_{x,y} a_x b_y g_x g_y \right) \cdot v \\
&= \sum_{x,y} a_x b_y (g_x g_y \cdot v) = \sum_{x,y} a_x b_y (g_x \cdot (g_y \cdot v)) \\
&= \sum_x a_x g_x \cdot \left(\sum_y b_y g_y v \right) = \sum_x a_x g_x \cdot \left(\sum_y b_y g_y \cdot v \right),
\end{aligned}$$

so $\cdot$ satisfies the module axioms.

2 Complete Reducibility

If W is a subrepresentation of a finite group G , then there exists a complementary invariant subspace W' of V such that $V = W \oplus W'$.

Suppose $W \leq V$, so V/W has vector space structure. Let $\{\bar{u}_\alpha\}_{\alpha \in J}$ be a basis of V/W . By axiom of choice, pick representatives $u_\alpha \in \bar{u}_\alpha$. Let $\{w_\beta\}_{\beta \in W}$ be a basis of W . Let $U = \langle u_\alpha \rangle_{\alpha \in J}$. I claim that $V = W \oplus U$.

Suppose $v \in V$. Write $\bar{v} = \sum \lambda_\alpha \bar{u}_\alpha = \overline{\sum \lambda_\alpha u_\alpha}$, so $u - \sum \lambda_\alpha u_\alpha \in W$. Write $u - \sum \lambda_\alpha u_\alpha = \sum \lambda_\beta w_\beta$, so $u \in \langle \{u_\alpha\} \cup \{w_\beta\} \rangle$.

I claim that $\{u_\alpha\} \cup \{w_\beta\}$ is a basis of V . Suppose $\sum \lambda_\alpha u_\alpha + \sum \lambda_\beta w_\beta = 0$. Then $\sum \lambda_\alpha \bar{u}_\alpha = 0$, so $\lambda_\alpha = 0$ for all α . Then $\sum \lambda_\beta w_\beta = 0$, so $\lambda_\beta = 0$ for all β .

Suppose $W \leq V$ is G -invariant and $V = W \oplus W'$. Is W' G -invariant?

Suppose $W \leq V$ is G -invariant. Define $G : V/W$ by $g(\bar{v}) = \overline{gv}$. Suppose $v - v' \in W$. Then $g(v - v') \in W$, so $gv - gv' \in W$, $G : V/W$ is well-defined.

We have $1(\bar{v}) = \overline{1v} = \bar{v}$ and $gh(\bar{v}) = \overline{(gh)v} = \overline{g(hv)} = g(\overline{hv}) = g(h(\bar{v}))$, so V/W is a rep.

$\text{Sps } \rho(g) : V \rightarrow V$ is G -linear. Then $\rho(g)\rho(h) = \rho(h)\rho(g)$, or $\rho(gh) = \rho(hg)$, or

$\rho([g, h]) = 1$, or $[g, h] \in \ker \rho$.

Let $S_3 \times \mathbb{C} \rightarrow \mathbb{C}$ be given by $(g, z) \mapsto \text{sgn}(g)z$. Note that $gh \cdot z = \text{sgn}(gh)z = \text{sgn}(g)\text{sgn}(h)z = g \cdot (h \cdot z)$. $1 \cdot z = 1z = z$. If g is fixed, the action is linear.

3 Characters

Sps $\rho_1 : G \rightarrow GL(V)$ is a rep. Define $\chi_V : G \rightarrow \mathbb{C}$ by $\chi(g) = \text{tr } \rho(g)$.

Claim: Let $\rho_1 : G \rightarrow GL(V_1), \rho_2 : G \rightarrow GL(V_2)$. Then $\chi_{V_1 \oplus V_2}(g) = \chi_{V_1}(g) + \chi_{V_2}(g)$.

Fix bases $\{v_{11}, v_{12}, \dots, v_{1n}\}$ and $\{v_{21}, v_{22}, \dots, v_{2m}\}$ of V_1, V_2 , resp. Then $[\rho(g)] = [\rho_1(g)] \oplus [\rho_2(g)]$, so $\chi_{V_1 \oplus V_2}(g) = \text{tr } [\rho(g)] = \text{tr } [\rho_1(g)] + \text{tr } [\rho_2(g)] = \chi_{V_1}(g) + \chi_{V_2}(g)$.

Compute

$$\det \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 0 & -1 \end{bmatrix} = 1(1) - 1(-3) + 1(2) = 6 \neq 0.$$

$$\chi_{V^{\oplus n}} = (2^n, 0, (-1)^n) = a(1, 1, 1) + b(1, -1, 1) + c(2, 0, -1).$$

Sps $G : V$. Let $V^G = \{v \in V : gv = v \forall g \in G\}$.

Define $\varphi = \frac{1}{|G|} \sum_{g \in G} g$. I claim that φ is G -linear.

Proof: We have

$$h \cdot \varphi(h^{-1} \cdot v) = \frac{1}{|G|} \sum_{g \in G} hgh^{-1}(v) = \frac{1}{|G|} \sum_{g \in G} gv = \varphi(v).$$

We have $g\varphi(v) = \varphi(gv) = \varphi(v)$. Sps $v \in V^G$. Then $\varphi(v) = \frac{1}{|G|} \sum_{g \in G} gv = \frac{1}{|G|} |G|v = v$. Hence, $\varphi : V \rightarrow V^G$ is a projection.

Sps $\{e_i\}_{i \in [n]}$ is a basis of V . Let $\{\epsilon_i\}$ be dual to $\{e_i\}$. I claim that $\chi_{V^*}(g) = \overline{\chi_V(g)}$ for all $g \in G$.

Proof: In general by argument of basis, we have $\chi_{V^*}(g) = \chi_V(g^{-1})$.

Note that G is finite. Fix a basis $\{e_1, \dots, e_n\}$ of V . Let $[g]$ denote the associated matrix. Note that $[g]^{|G|} = 1$, so $[g]$ is diagonalizable. In particular, each eigenvalue of $[g]$ is an n th root of unity. It follows that $\chi_V(g^{-1}) = \overline{\chi_V(g)}$ since

for z with $|z| = 1$, we have $z^{-1} = \bar{z}$.

If V, W are irred, then

$$\dim \text{Hom}(V, W) = \begin{cases} 1 & \text{if } V \cong W \\ 0 & \text{if } V \not\cong W. \end{cases}$$

We have $\chi_{\text{Hom}(V, W)}(g) = \overline{\chi_V(g)} \cdot \chi_W(g)$.

Write $V = \bigoplus_{i=1}^n V_i^{\oplus a_i}$. Then $\chi_V = \sum_{i=1}^n a_i \chi_{V_i}$ and $(\chi_V, \chi_V) = \sum_{i=1}^n a_i^2 = 1 \rightarrow a_i = 1$ for a unique $i \in [n]$.

Consider the regular representation R of G . We have

$$\chi_R(g) = \begin{cases} 0 & \text{if } g \neq 1 \\ |G| & \text{if } g = 1. \end{cases}$$

For $G \neq 1$, we see that R is not irreducible.

Write $R = \bigoplus V_i^{\oplus a_i}$. Then

$$(\chi_R, \chi_{V_i}) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_R(g)} \chi_{V_i}(g) = \frac{1}{|G|} |G| \dim V_i = \dim V_i.$$

Hence, the irred V_i appears $\dim V_i$ times in R . As a corollary, $|G| = \dim R = \sum_{i=1}^r (\dim V_i)^2$.

4 More Projection Formulas, Consequences

Let V be an irrep, W a rep. Consider $z_{\bar{V}} = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} g$. Since χ_V is a class function, $z_{\bar{V}} \in Z(\mathbb{C}[G])$. Note that $\rho_W(z_{\bar{V}}) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \rho_W(g)$ is a G -linear map from W to W .

Let V, W be irreps. I claim that

$$\frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \chi_W(g) = \begin{cases} 0 & \text{if } V \not\cong W \\ 1 & \text{if } V \cong W. \end{cases}$$

Let $\rho : G \rightarrow GL(V)$ be a rep. I claim that $\text{pr} : V \rightarrow V$ given by

$$\text{pr}(v) = \frac{1}{|G|} \sum_{g \in G} \rho(g)(v)$$

is a G -linear projection of V onto V^G .

Proof: Sps $v \in V^G$. Then

$$\text{pr}(v) = \frac{1}{|G|} \sum_{g \in G} \rho(g)(v) = \frac{1}{|G|} \sum_{g \in G} v = \frac{|G|}{|G|} v = v.$$

Now suppose $g' \in G$. We have

$$\begin{aligned} \rho(g')(\text{pr}(v)) &= \rho(g') \left(\frac{1}{|G|} \sum_{g \in G} \rho(g)(v) \right) = \frac{1}{|G|} \sum_{g \in G} \rho(g')(\rho(g)(v)) \\ &= \frac{1}{|G|} \sum_{g \in G} \rho(g'g)(v) = \frac{1}{|G|} \sum_{g \in G} \rho(g)(v) = \text{pr}(v), \end{aligned}$$

so $\text{pr}(V) \leq V^G$. Finally,

$$\text{pr}(\rho(g')(v)) = \frac{1}{|G|} \sum_{g \in G} \rho(g)(\rho(g')(v)) = \frac{1}{|G|} \sum_{g \in G} \rho(gg')(v) = \text{pr}(v),$$

so pr is G -linear.

Let V, W be irreps. Note

$$\dim \text{Hom}_G(V, W) = \begin{cases} 0 & \text{if } V \not\cong W, \\ 1 & \text{if } V \cong W. \end{cases}$$

Identify $\text{Hom}_G(V, W) = (V^* \otimes W)^G$. Note $\dim V^G = \chi_{\text{pr}}$.

Let $\text{pr} : V \rightarrow V$ be the projection of V onto V^G . Let $e_1, e_2, \dots, e_m$ be a basis of V^G . Extend $\{e_i\}_{i \in [m]}$ to $\{e_i\}_{i \in [n]}$, where $n = \dim V$. Note that $\text{pr}(e_i) = e_i$ for $i \in [m]$. If $i > m$, write $\text{pr}(e_i) = \sum_{i \in [n]} \lambda_i e_i$. Since $\text{pr}(e_i) \in V^G$, we have $\lambda_i = 0$ for $i > m$. Then $\text{tr pr} = \dim V^G$.

Then

$$\text{tr}(\text{pr}) = \frac{1}{|G|} \sum_{g \in G} \text{tr}(\rho(g)) = \frac{1}{|G|} \chi_V(g).$$

We compute

$$\text{tr}(\text{pr}) = \frac{1}{|G|} \sum_{g \in G} \chi_{V^* \otimes W} = \frac{1}{|G|} \overline{\chi_V(g)} \chi_W(g).$$

Let V, W be irreps of G . Let $z_{\overline{V}} = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} g \in Z(\mathbb{C}[G])$. Then

$$\rho_W(z_{\overline{V}}) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \rho_W(g)$$

is a G -linear map $W \rightarrow W$. Since W is irrep, $\rho_W(z_{\overline{V}})$ acts as a scalar. We compute $\text{tr } \rho_W(z_{\overline{V}}) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \chi_W(g) = 1$ if $V \cong W$ and 0 otherwise. Hence, $\rho_W(z_{\overline{W}})$ acts by multiplication by $1/\dim W$ if $W \cong V$ and by 0 otherwise.

Let $f : G \rightarrow \mathbb{C}$ be a class function. I wish to show

$$f = \sum_{i \in [r]} \lambda_i \chi_{V_i}$$

for some $\lambda_i \in \mathbb{C}$. Consider the assoc $z = \sum_{g \in G} f(g)g \in Z(\mathbb{C}[G])$.

Consider $z = \sum_{i \in [r]} \lambda_i z_{\overline{V_i}}$. Then

$$\begin{aligned} \sum_{g \in G} f(g)g &= \frac{1}{|G|} \sum_{i \in [r]} \sum_{g \in G} \overline{\chi_{V_i}(g)} g = \frac{1}{|G|} \sum_{g \in G} \sum_{i \in [r]} \overline{\chi_{V_i}(g)} g \rightarrow \\ f(g) &= \frac{1}{|G|} \sum_{i \in [r]} \lambda_i \overline{\chi_{V_i}(g)} \end{aligned}$$

Claim: If $\{V_\mu\}$ is the complete collection of irreps, then so is $\{V_\mu^*\}$.

Proof: Let $\{e_i\}, \{e_i^*\}, \{e_i^{**}\}$ be bases of V, V^*, V^{**} . Let $[f_j]_{j \in [n]} = [e_j]_{j \in [n]} C$, so $f_j = \sum_{i \in [n]} c_{ij} e_i$.

Let $z \in Z(\mathbb{C}[G])$. I claim that $\rho_V(z) \in \text{Hom}_G(V, V)$.

Proof:

$$\begin{aligned} \rho_V(z)(g' \cdot v) &= \frac{1}{|G|} \sum_{g \in G} f(g)g \cdot (g' \cdot v) = \frac{1}{|G|} \sum_{g \in G} f(g)gg' \cdot v \\ &= \frac{1}{|G|} \sum_{g \in G} f(g)g'g \cdot v = g' \cdot \rho_V(z)(v), \end{aligned}$$

as desired.

Let $\Psi : V \rightarrow V^{**}$ be given by $\Psi(v)(\varphi) = \varphi(v)$ for all $v \in V, \varphi \in V^*$. Note

$$\begin{aligned} \Psi(v+w)(\varphi) &= \varphi(v+w) = \varphi(v) + \varphi(w) = \Psi(v)(\varphi) + \Psi(w)(\varphi) \\ \Psi(\lambda v)(\varphi) &= \varphi(\lambda v) = \lambda \varphi(v) = \lambda \Psi(v)(\varphi), \end{aligned}$$

so Ψ is linear. Suppose $\Psi(v) = \Psi(w)$. Then $\varphi(v) = \varphi(w)$ for all $\varphi \in V^*$. Fix a basis $\{\epsilon_i\}$ of V^* . Then $\epsilon_i(v) = \epsilon_i(w)$ for all $i \in [n]$, so $v = w$. Hence, Ψ is injective.

Since V, V^{**} are both dimension n , Ψ is a linear isomorphism.

Compute $\Psi(e_i)(e_j^*) = \epsilon_j^*(e_i) = \delta_{ij}$

Let $\alpha : G \rightarrow \mathbb{C}$ be any function and $G : V$. Then $\varphi_{\alpha,V} = \sum_{g \in G} \alpha(g)g$ is G -linear iff α is a class function for all V .

Proof: Suppose α is a class function. Then

$$\begin{aligned} \varphi_{\alpha,V}(g' \cdot v) &= \sum_{g \in G} \alpha(g)g \cdot (g' \cdot v) = \sum_{g \in G} \alpha(g)gg' \cdot v \\ &= \sum_{g \in G} \alpha(g'g(g')^{-1})g'g \cdot v = \sum_{g \in G} \alpha(g)g'g \cdot v \\ &= g' \cdot \left(\sum_{g \in G} \alpha(g)g \cdot v \right) = g' \cdot \varphi_{\alpha,V}(v), \end{aligned}$$

so $\varphi_{\alpha,V}$ is G -linear.

Now suppose $\varphi_{\alpha,V}$ is G -linear for all V . Take $V = \mathbb{C}[G]$. Then

$$\begin{aligned} g' \cdot \varphi_{\alpha,V}(v) &= \varphi_{\alpha,V}(g' \cdot v) \rightarrow \\ \sum_{g \in G} \alpha(g)g'g \cdot v &= \sum_{g \in G} \alpha(g)gg' \cdot v \rightarrow \\ \sum_{g \in G} \alpha(g)g'g &= \sum_{g \in G} \alpha(g)gg' \rightarrow \\ \alpha(g) &= \alpha(g'g(g')^{-1}), \end{aligned}$$

so α is a class function.

Let $f : G \rightarrow \mathbb{C}$ be a class function. Consider

$$f = \sum_{i \in [r]} a_i \chi_i \rightarrow \langle \chi_i, f \rangle = a_i,$$

so I claim that

$$f = \sum_{i \in [r]} \langle \chi_i, f \rangle \chi_i.$$

Write $V = \bigoplus_{i \in [r]} V_i^{\oplus a_i}$. Consider φ_{f,V_i^*} , which is G -linear. Since $\varphi_{f,V_i^*} : V_i^* \rightarrow V_i^*$, φ_{f,V_i^*} acts as multiplication by a scalar, and this scalar is $1/\dim V_i$ of the trace of φ_{f,V_i^*} wrt any basis.

We compute

$$\text{tr } \varphi_{f,V_j^*} = \sum_{i \in [r]} \langle \chi_i, f \rangle \text{tr } \varphi_{\chi_i,V_j^*}$$

and

$$\text{tr } \varphi_{\chi_i, V_j^*} = \text{tr } \sum_{g \in G} \chi_i(g)g = \sum_{g \in G} \chi_i(g)\overline{\chi_j(g)} = |G|\langle \chi_j, \chi_i \rangle,$$

so φ_{χ_i, V_j^*} acts as multiplication by 0 if $i \neq j$ and by multiplication by $|G|/\dim V_j$ otherwise.

In particular, φ_{f, V_j^*} acts by multiplication by $\langle \chi_j, f \rangle |G|/\dim V_j$. Other side:

$$\text{tr } \varphi_{f, V_j^*} = \text{tr } \sum_{g \in G} f(g)g = \sum_{g \in G} f(g)\overline{\chi_j(g)} = |G|\langle \chi_j, f \rangle,$$

so $\text{tr } \varphi_{f, V_j^*}$ acts by multiplication by the same constant $\langle \chi_j, f \rangle |G|/\dim V_j$. Write $V = \bigoplus_{i \in [r]} (V_i^*)^{\oplus a_i}$. In particular, the action of the LHS and RHS coincide for $V = \mathbb{C}[G]$, so LHS = RHS.

$$\mathbb{C}_{class} \hookrightarrow \text{Hom}_G(\mathbb{C}[G], \mathbb{C}[G]).$$

Define $\varphi : V \rightarrow V^{**}$ by $\varphi(v)(w^*) = w^*(v)$. We have

$$\begin{aligned} \varphi(v_1 + v_2)(w^*) &= w^*(v_1 + v_2) = w^*(v_1) + w^*(v_2) = \varphi(v_1)(w^*) + \varphi(v_2)(w^*) \\ \varphi(\lambda v)(w^*) &= w^*(\lambda v) = \lambda w^*(v) = \lambda \varphi(v)(w^*), \end{aligned}$$

so φ is linear.

Suppose $\varphi(v_1) = \varphi(v_2)$. Let $\{e_i\}$ be a basis of V and $\{e_i^*\}$ be dual in V^* . Note $e_i^*(v_1) = e_i^*(v_2)$ for all $i \in [n]$, so $v_1 = v_2$, so φ is injective.

Let $e_i \rightarrow e_i^* \rightarrow e_i^{**}$. We have $e_i^{**}(e_j^*) = \delta_{ij} = \varphi(e_i)(e_j^*)$, so $\varphi(e_i) = e_i^{**}$ for all $i \in [n]$.

Sps $G : V$. Then $G : V^*$ and $G : V^{**}$. $(g \cdot v^{**})(w^*) = v^{**}g^{-1}(w^*) = v^{**}(w^*g) = (w^*g)(v) = w^*(g \cdot v)$. $(g \cdot v)^{**}(w^*) = w^*(g \cdot v)$, so the natural map $V \rightarrow V^{**}$ is an isomorphism of representations.

$$V^* \cong W^* \rightarrow V^{**} \cong W^{**} \rightarrow V \cong W.$$